



PHILIPS



CDI660/CDI670

Technical documentation

1. INTRODUCTION

1.1 Introduction.....	3
1.2 Front of player.....	4
1.3 Rear of player.....	4
1.4 Remote control.....	5
1.5 General technical characteristics	5

2. PORT SPECIFICATIONS

2.1 Introduction.....	6
2.2 Serial ports.....	6
2.2.1 Input 1	6
2.2.2 Input 2	6
2.2.3 Serial I/O port	6
2.2.4 Port #1 & Port #2.....	7
2.3 Parallel port #3.....	7
2.4 AT Keyboard port #4	7

3. INPUT DEVICES

3.1 Introduction.....	8
3.2 Device identification	8
3.2.1 Name convention.....	8
3.2.2 Device ID sequence.....	8
3.2.3 Device ID codes.....	8
3.2.4 Device ID timing.....	8
3.3 MOUSE/TRACKERBALL specification	9
3.3.1 Data rate	9
3.3.2 Data flow	9
3.3.3 Data format.....	9
3.3.4.Optional byte.....	9
3.4 JOYSTICK specification (touchpad/gamepad)	10
3.4.1 Data rate	10
3.4.2 Data flow	10
3.4.3 Data format.....	10
3.4.4 Green Book Specifications	10
3.4.5 Digital and analog Joystick.....	10
3.4.6 Optional byte 3	10
3.5 GRAPHIC TABLET/TOUCH SCREEN specification.....	11
3.5.1 Data rate	11
3.5.2 Data flow	11
3.5.3 Data format.....	11
3.6 CD-i KEYBOARD specification.....	11
3.6.1 Data rate	11
3.6.2 Data flow	11
3.6.3 Data format.....	11

3.6.4 Implementation of a USA English version keyboard	12
3.6.5 Implementation of an extended keyboard	13
3.7 AT KEYBOARD SPECIFICATIONS.....	14
3.7.1 Data flow	14
3.7.2 Data format.....	14
3.8 PCK TABLE	15
3.8.1 PCK's.....	15
3.8.2 Definition of 'green' PCK's	15
3.8.3 Definition of 'non green' PCK's.....	15

4. ZIP DISK SPECIFICATION

4.1 Introduction.....	16
4.2 Hardware specifications.....	16
4.3 Software specifications.....	16

5. SYSTEM CONFIGURATION

5.1 Introduction	17
5.2 Updating the CSD file depending on devices present....	17
5.3 Load application.....	17
5.4 Load system modules.....	18

6. PLAYER STATISTICS

7. GROUP WORKING

7.1 Introduction.....	19
7.2 Definitions	19
7.3 Preconditions.....	19
7.4 Network configuration.....	19
7.5 CD-i network protocol.....	19
7.6 Layer 1: physical link layer.....	19
7.7 Layer 2: data link layer	20
7.8 Layer 3: network layer.....	20
7.9 Layer 4: transport layer.....	20
7.10 Layer 5: session layer	20
7.11 Summary	20

8. KEYDISK FUNCTIONS

8.1 Locking features.....	21
8.2 Lock functions.....	21
8.3 Secondary locking functions	21
8.4 CD-DA lockout.....	21
8.5 Other key disk functions	21

I.1 Introduction

This is the technical documentation for the CDI660 and CDI670 serie of CD-i players. This documentation is especially made to be able to support development of sophisticated CD-i applications.

As this documentation describes different player models, some interfaces and related funtion descriptions are not available on alle models.

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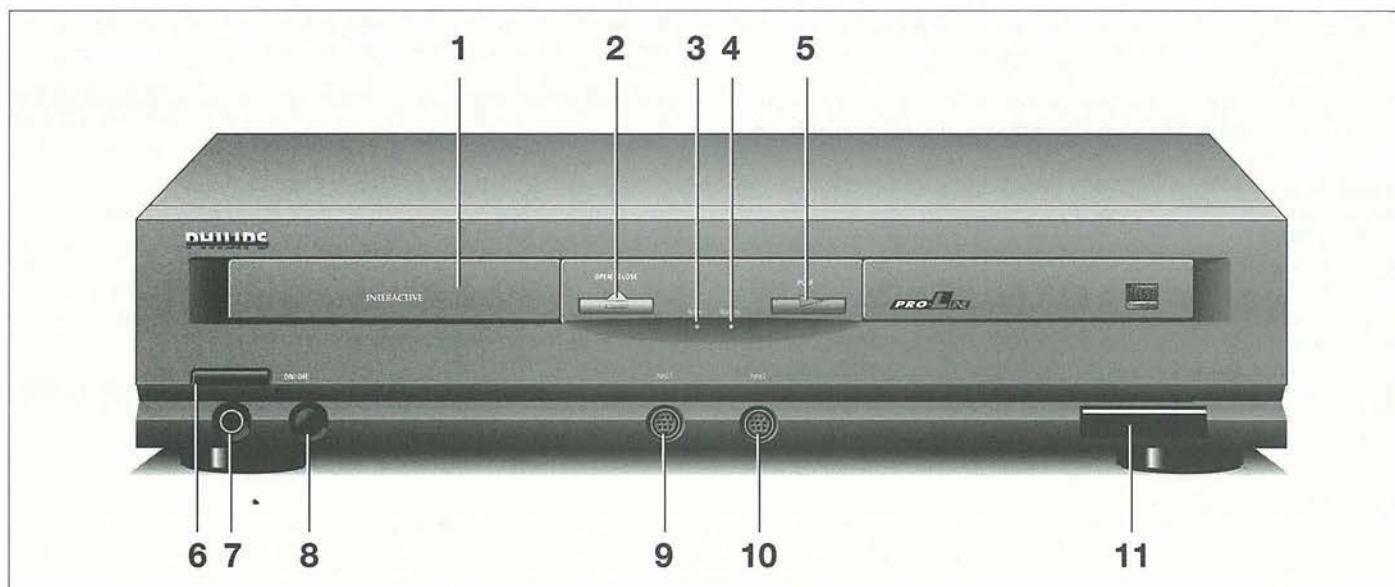
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REGISTERED NAMES

Registered names used in this document are:
Amikey-2 (American Megatrends), Centronics, IBM, Iomega, Philips, Super I/O (Standard Microsystems Corporation), Texas (Texas Instruments), Zip (Iomega).

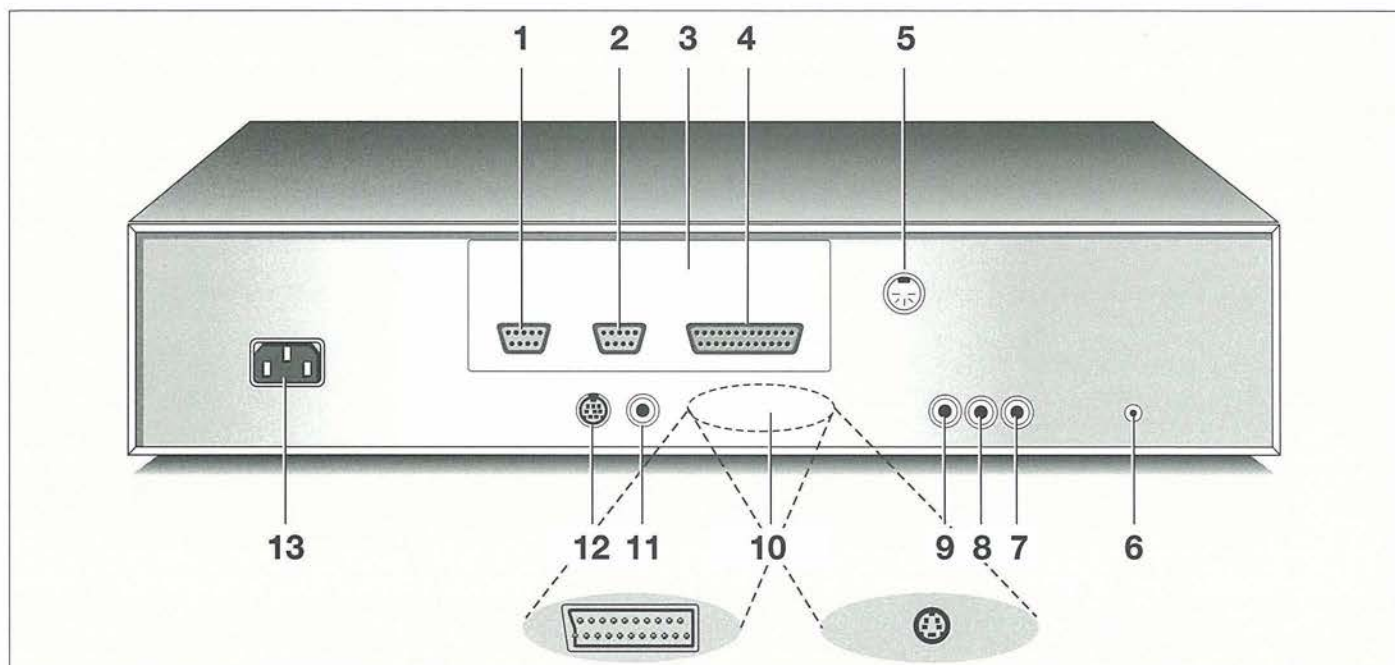
I. INTRODUCTION

I.2 Front of player



- | | |
|---------------------------------|---|
| 1 Disc loading tray. | 7 Headphone socket. |
| 2 Open/close button. | 8 Headphone volume control. |
| 3 'CD-i busy' indicator (green) | 9 Input 1: pointing device 1. |
| 4 'On/Off' indicator (red) | 10 Input 2: pointing device 2. |
| 5 Play button. | 11 IR-receiver eye with 'Active' indicator (red). |
| 6 Power On/Off button. | |

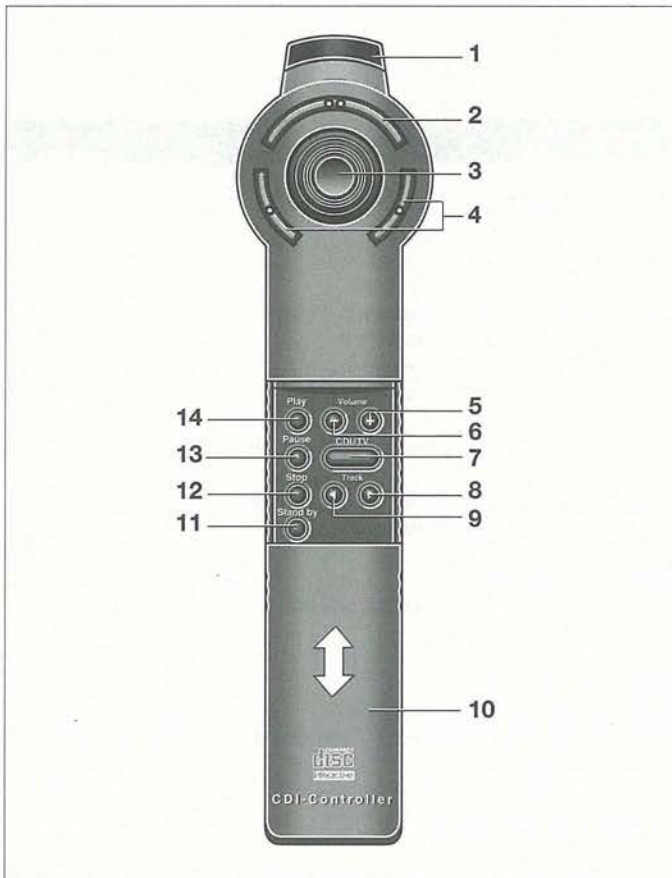
I.3 Rear of player



- | | |
|--|---------------------------------|
| 1 Port #1: RS232 serial port.* | 8 Audio L-channel connector. |
| 2 Port #2: RS232 serial port.* | 9 Audio R-channel connector. |
| 3 Cover plate for extensions.* | 10 A/V Euroconnector.* |
| 4 Port #3: parallel port.* | 10 Y/C (S.-Video) connector.* |
| 5 Port #4: AT-keyboard connector.* | 11 Digital output connector. |
| 6 +5V/Audio out (for optional RF Modulator). | 12 Serial I/O port. |
| 7 CVBS/Video connector. | 13 Power connector (AC socket). |

* Available on some players only.

I.4 Remote control



- 1 Infrared remote transmitter
- 2 Action button.
- 3 Thumbpad.
- 4 Action buttons (for left or right handed operation).
- 5 Volume up button.
- 6 Volume down button.
- 7 CD-i/TV button*.
- 8 Next track**.
- 9 Previous track**.
- 10 Sliding Cover.
- 11 Standby*.
- 12 Stop button**.
- 13 Pause button**.
- 14 Play button**.

* Not functional on some CD-i player models.

** Only functional for CD-Audio and some CD-i applications (see also 3.8).

I.5 General technical characteristics

AUDIO:

- L+R (Cinch connector) 2 V rms typical 10 KOhm.
- Headphone (6 mm Headphone socket).
8 Ohm - 2 KOhm, 30 mW at 32 Ohm.
- Manual volume control.

VIDEO:

NTSC (525 lines) or PAL (625 lines) video out:

- RGB (A/V Euroconnector) 0.7 Vp-p into *
75 Ohm (0.3 Vp-p into 75 Ohm sync).
- Y/C (Y/C connector) 1 Vp-p into 75 Ohm *
- CVBS (Cinch connector) 1 Vp-p into 75 Ohm.

CONTROL PORTS:

- Input 1 (miniDIN 8-pin connector) for pointing device.
- Input 2 (miniDIN 8-pin connector) for pointing device or CD-i keyboard.
- Port #1 (D-sub 9 connector) for RS232-C based peripherals. *
- Port #2 (D-sub 9 connector) for RS232-C based peripherals. *
- Parallel port: *
 - connector: 25 pin D-sub connector.
 - standard: Centronics compatible.
 - function: serves as a printer interface or external disk drive interface.
- AT keyboard connector (PC-AT type). *
- Serial I/O port (miniDIN 8 pin connector) for RS based peripherals (TTL level) like modem.

REMOTE CONTROL: *

- RC6 compatible.

MEMORY:

- 2 x 512 Kb base case RAM + 1.5 Mb (0.5 Mb MPEG + 1.0 Mb RAM).
- 32 Kb non volatile RAM.
- 2 x 512 Kb ROM.

CAUTION

(FOR APPLICATION/TITLE DEVELOPMENT)

STANDARD CD-i SYSTEMS NORMALLY HAVE SLIGHTLY MORE RAM THAN THE 2X480K FREE AVAILABLE MEMORY. TO AVOID PROBLEMS DURING APPLICATION/TITLE DEVELOPMENT IT IS ADVISABLE TO USE ONLY COLORED MEMORY.

CPU:

68070 at 15 Mhz.

* Not available on some CD-i player models.

2. PORT SPECIFICATIONS

2.1 Introduction

This chapter gives the specification of the ports on a CD-i player. Each port is described in detail below. This chapter only gives the hardware specification. The software specification is given in chapter 3 'Input devices'.

2.2 Serial ports

2.2.1 INPUT 1

- Connector type: 8 pin miniDIN (female).
- Location: front.
- Usage: for standard CD-i pointing devices (TTL).
- Internal device: Slave proc.(68HC0518, SCI).
- Buffers: receiver: 1 byte.
Receive shift register not included.
- Baudrate: 1200.
- Data format: for pointing devices: 1 startbit, 7 databits, 2 stopbits, no parity.
- Handshake: hardware: only RTS.
- Signal levels: TTL compatible; input pins also accept RS levels.

Signals & Pinning:

Pin	Signal	Description	Inp/Outp.
1	n.c.	Not Connected	
2	RXD	Received Data	Input
3	TXD	Transmitted Data (not used)	Output
4	n.c.	Not Connected	
5	GND	Signal Ground	
6	n.c.	Not Connected	
7	RTS	Request To Send	Output
8	+5V	Power supply for the device	

Input: this is an input signal for the CD-i player.

Output: this is an output signal from the CD-i player.

RATINGS: +5V: +/-10% max. 100mA.

TTL level definitions are shown below:

Signal level (TTL)	< + 0.8 V	> +2V
Binary state (for DATA)	1	0
Signal Condition	Mark	Space
Control & Timing Function	OFF (Negated)	ON(Asserted).

2.2.2 INPUT 2

- Connector type: 8 pin miniDIN (female).
- Location: front.
- Usage: for standard CD-i pointing devices and CD-i keyboard (TTL).
- Internal device: Slave proc.(68HC0518, SCI).
- Buffers: receiver: 1 byte.
Receive shift register not included.
- Baudrate: 1200.
- Data format: for pointing devices: 1 startbit, 7 databits, 2 stopbits, no parity.
For CD-i keyboards: 1 startbit, 8 databits, 1 stopbit, no parity.
- Handshake: hardware: only RTS.
- Signal levels: TTL compatible; input pins also accept RS levels.

Signals & Pinning:

Pin	Signal	Description	Inp/Outp.
1	n.c.	Not Connected	
2	RXD	Received Data	Input
3	n.c.	Not Connected	
4	n.c.	Not Connected	
5	GND	Signal Ground	
6	n.c.	Not Connected	
7	RTS	Request To Send	Output
8	+5V	Power supply for the device	

Input: this is an input signal for the CD-i player.

Output: this is an output signal from the CD-i player.

RATINGS: +5V: +/-10% max. 100mA.

TTL level definitions are shown below:

Signal level (TTL)	< + 0.8 V	> +2V
Binary state (for DATA)	1	0
Signal Condition	Mark	Space
Control & Timing Function	OFF (Negated)	ON(Asserted).

2.2.3 SERIAL I/O PORT

- Connector type: 8 pin miniDIN (female).
- Location: rear.
- Usage: for RS based peripherals, like modems.
- Internal device: on chip UART (SCC68070).
- Buffers: transmitter: 1 byte receiver: 1 byte.
Transmit & receive shift register not included.
- Baudrate: 9600.
- Data format: character length: 7 or 8 bits.
Stopbits length: 1 or 2 bits.
Parity: odd, even or no.
- Handshake: hardware:
2 control lines; RTS and CTS
- Software: XON/XOFF.

Signals & Pinning:

Pin	Signal	Description	Inp/Outp.
1	n.c.	Not Connected	
2	RXD	Received Data	Input
3	TXD	Transmitted Data	Output
4	n.c.	Not Connected	
5	GND	Signal Ground	
6	CTS	Clear To Send	Input
7	RTS	Request To Send	Output
8	+5V	Power supply for the device	

Input: this is an input signal for the CD-i player.

Output: this is an output signal from the CD-i player.

RATINGS: +5V: +/-10% max. 100mA.

TTL level definitions are shown below:

Signal level (TTL)	< + 0.8 V	> +2V
Binary state (for DATA)	1	0
Signal Condition	Mark	Space
Control & Timing Function	OFF (Negated)	ON(Asserted).

2. PORT SPECIFICATIONS

2.2.4 PORT #1 & PORT #2

- Connector type: 9 pin sub-D (male).
- Location: rear.
- Usage: for RS232 devices/general purpose serial communic.
- Internal device:
PORT #1 = FDC37C665GT and Texas LBC 241.
PORT #2 = FDC37C665GT and Texas LBC 241.
- Buffers: 16 byte, transmit and receive FIFO.
- Baudrates: 50, 75, 110, 134.5, 150, 300, 600, 1200, 1800, 2400, 4800, 7200, 9600, 19200 and 38400 up to 115.2 K.
Independently programmable baudrate for receiver and transmitter.
- Dataformats: character length: 5, 6, 7 or 8 bits.
Stopbit length: 1 or 2 stopbits.
Parity: odd, even or no.
- Handshake:
hardware: CTS, RTS, DTR, DSR, DCD.
Software: XON/XOFF.
Specifications according to EIA/TIA RS-232E and CCITT V.28.
- These inputs/outputs are full RS compatible.
- Signal levels: + and - 12V

Signals & Pinning:

Pin	Signal	Description	Inp/Outp.
1	DCD	Data Carrier Detect	Input
2	RXD	Received Data	Input
3	TXD	Transmit Data	Output
4	DTR	Data Terminal Ready	Output
5	GND	Signal Ground	
6	DSR	Data Set Ready	Input
7	RTS	Request To Send	Output
8	CTS	Clear To Send	Input
9	RI	Ring Indicator	Input

Input: this is an input signal for the CD-i player.

Output: this is an output signal from the CD-i player.

Connector pinout = IBM (Registered Name) PC-AT pinout compatible.

2.3 Parallel port #3

- Connector type: DB-25 (fem.).
- Location: rear.
- Usage: for printers and general purpose parallel I/O.
Bi-directional 8 bit PC/Centronics compatible.
- Signal levels: input and outputs are TTL levels.

Signals & Pinning:

Pin	Signal	I/O	Pin	Signal	I/O
1	STROBE*	O	14	AUTOFDXT*	O
2	D0	I/O	15	ERROR*	I
3	D1	I/O	16	INIT* (reset)	O
4	D2	I/O	17	SEL* (select in)	O
5	D3	I/O	18	GROUND	-
6	D4	I/O	19	GROUND	-
7	D5	I/O	20	GROUND	-
8	D6	I/O	21	GROUND	-
9	D7	I/O	22	GROUND	-
10	ACK*	I	23	GROUND	-
11	BUSY	I	24	GROUND	-
12	PE (paper empty)	I	25	GROUND	-
13	SEL (on-line)	I			

* Low = active

2.4 AT Keyboard port #4

- Connector type: 5 pin DIN.
- Location: rear.
- Usage: for PC/AT compatible keyboard or compatible peripherals.
- Internal device: AMIKEY-2 (Registered Name)
- Signal levels: TTL.

Signals & Pinning:

pin	signal
1	clock
2	data
3	n.c.
4	GND
5	+5V (300 mA max)

3. INPUT DEVICES

3.1 Introduction

This chapter describes some pointing devices, keyboards and modems for a CD-i player.

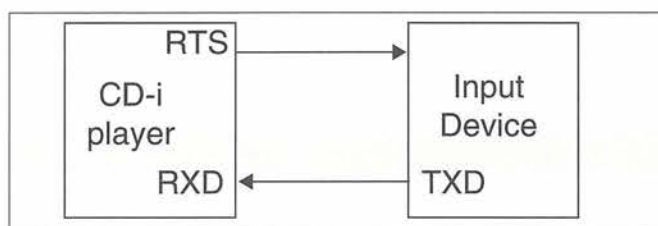
The most important items will be discussed for each device.

- Data rate.
- Data flow.
- Data format.

3.2 Device Identification

3.2.1 Name convention

For identification of pointing devices, keyboards and modems, there are only two signals which will carry data.



The first is the RTS signal and is always an output for the CD-i player. The second is the RXD signal and is always an input for the CD-i player.

- RXD : carries the data from pointing device, keyboard or modem to player.
- RTS : is used for device identification.

To identify the input device that is connected to the CD-i player, each input device will transmit its own device (ID) byte on request or after power on.

3.2.2 Device ID sequence

- 1 Negate RTS: Current data transmission to the CD-i player is stopped.
- 2 Assert RTS.
- 3 The device will send its 1 byte device ID.
- 4 Sending a complete data packet is recommended after the identification byte to determine the state of the buttons at start-up.

NOTE

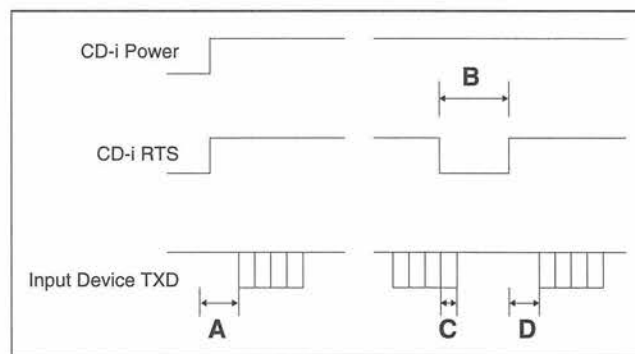
- The CD-i keyboard sends its 1 byte device ID, and clears its buffer.
- Data transmission to the CD-i player can continue. The first byte sent to the CD-i player should be the header byte of the device data packet (the first byte of a data packet d7 = 1).

3.2.3 Device ID codes

	d6	d5	d4	d3	d2	d1	d0	
Mouse / Trackerball:	1	0	0	1	1	0	1	= ascii char. 'M' (0x4D)
Joystick :	1	0	0	1	0	1	0	= ascii char. 'J' (0x4A)
Graph.tab.:	1	0	1	0	1	0	0	= ascii char. 'T' (0x54)
Touch scr.:	1	0	1	0	0	1	1	= ascii char. 'S' (0x53)
Modem :	0	1	1	0	0	0	0	= ascii char. 'O' (0x30)
Ext.keyb.*:	1	0	0	1	0	1	1	= ascii char 'K' (0x4B)
keyboard*:	1	0	1	1	0	0	0	= ascii char 'X' (0x58)

* = CDi-Keyboards

3.2.4 Device ID timing



- A: device power-on to ID byte timing:
Min. 100ms
Max. 500ms.
- B: negate RTS time:
Min. 10ms
Max. irrelevant.
- C: Continued data transmission time when RTS is negated:
Min. 0ms
Max. 1 byte time (1200 baud = 8.3ms, 9600 baud = 1ms).
- D: assert RTS to ID byte timing when device is powered:
Min. 1 bit time (1200 baud = 0.8 ms, 9600 baud = 0.1 ms).
Max. 10 ms.

NOTE

- Using the optional key disk an extended (max. 1.2 sec.) power-on to ID byte timing can be realised.

3.3 MOUSE/TRACKERBALL specification

3.3.1 Data rate

Baudrate: 1200 baud.

3.3.2 Data flow

During normal operation (RTS asserted) :

- Data is received by the CD-i player via RXD.
- Every time there is a mouse movement or a button state change, the three byte mouse data is transmitted from mouse to CD-i player.
- When the mouse is not moved and the buttons do not change state (they stay as they are, pressed or released), the mouse should not send any 3 byte mouse data to the CD-i player.
- An on going data packet transmission should not be interrupted by a button state change or a mouse position change.

3.3.3 Data format

- 1 startbit
- 7 databits (LSB first)
- 2 stopbits

	d6	d5	d4	d3	d2	d1	d0
Byte 0	1	bt1	bt2	Y7	Y6	X7	X6
Byte 1	0	X5	X4	X3	X2	X1	X0
Byte 2	0	Y5	Y4	Y3	Y2	Y1	Y0
Byte 3	optional						

- bt1 = button1: 1 = pressed 0 = not pressed.
- bt2 = button2: 1 = pressed 0 = not pressed.
- X7..X0:
 - 8 bit data horizontal movement.
 - 2's complement.
 - left direction = negative (\$FF..\$80).
 - right direction = positive (\$01..\$7F).
- Y7..Y0:
 - 8 bit data vertical movement.
 - 2's complement.
 - up direction = negative (\$FF..\$80).
 - down direction = positive (\$01..\$7F).

A databyte = \$00 indicates that there was no movement in that direction.

3.3.4 Optional byte 3

- This optional byte indicates a Player control Key (PCK). PCK (6 bit) is applicable (see 3.8 PCK table for values).
- Note : d6 is 0.

3. INPUT DEVICES

3.4 JOYSTICK specification (Touchpad / Gamepad)

3.4.1 Data rate

Baudrate: 1200 baud.

3.4.2 Data flow

- Data rate: the maximum possible data rate from joystick to CD-i player is 10 bits at 1200 baud or 1 byte every 8.3ms. A complete data packet takes $3 * 8.3 = 25\text{ms}$. Therefore, the maximum data rate is 40 packets per second.
- Pixels in X and Y: normal resolution = $384 * 280$.
- Speed: depends on how fast the cursor must move over the screen.

General formula: $D = NP / (SP * DR)$

D: Data value at maximum speed.

NP: Number of pixels.

SP: Cursor speed (in seconds).

DR: Data rate (in datapackets per second).

During normal operation (RTS asserted) :

- Data is received by the CD-i player via RXD.
- A 3 byte data packet is only sent to the CD-i player continuously when the stick is moved out of its center position.
- When the stick is released (center position) and the buttons do not change state (they stay as they are, pressed or released), the joystick should not send any 3 byte joystick data to the CD-i player.
- Every time a button changes state, a 3 byte data packet is sent to the CD-i player.
- An on going data packet transmission should not be interrupted by a button change or a stick position change.

3.4.3 Data format

- 1 startbit
- 7 databits
- 2 stopbits

	d6	d5	d4	d3	d2	d1	d0
Byte 0	1	bt1	bt2	Y7	Y6	X7	X6
Byte 1	0	X5	X4	X3	X2	X1	X0
Byte 2	0	Y5	Y4	Y3	Y2	Y1	Y0
Byte 3 optional							

- bt1 = button1: 1 = pressed 0 = not pressed.
- bt2 = button2: 1 = pressed 0 = not pressed.
- X7..X0:
 - 8 bit data horizontal movement.
 - 2's complement.
 - left direction = negative (\$FF..\$80).
 - right direction = positive (\$01..\$7F).
- Y7..Y0:
 - 8 bit data vertical movement.
 - 2's complement.
 - up direction = negative (\$FF..\$80).
 - down direction = positive (\$01..\$7F).

A databyte = \$00 indicates that there was no movement in that direction.

3.4.4 Green Book specifications

- A minimum of 16 directions equally spaced around the full 360 degrees are needed. (FFGB:A VII.2-17).
- A pointing device must be able to point every pixel on the screen in normal resolution.
The pointing device always sends high resolution coördinates.

3.4.5 Digital and analog joystick

The following examples show some parameters that should be considered when developing a digital or an analog joystick.

Joystick DIGITAL type (3 speeds)

- It should be possible to pinpoint every pixel on the screen so that when the stick is in the first position (from center), the X and/or Y data sent should be +1 or -1. (This is the slowest speed).
- Assuming we want to move the cursor over the screen in about 1 second.
 - 720 pixels in 1 second.
 - The joystick transmits data packets at a rate of 40 per second.
 - The value (+ or -) that should be transmitted is $720/40$ or appr. \$12.
 - This could be the value for the outermost position in X or Y direction. (This is the highest speed).
- A value for the middle position could be \$09. This is the middle speed.
- For the digital type joystick this would give the following values:
 - Positive: \$01,\$09 and \$12.
 - Negative: \$FF,\$F7 and \$EE.

Joystick ANALOG type

- Assume again that we want to move over the screen in about 1 second. The maximum value for X and Y (in + or - direction) should be \$12.
- The center position should give \$00.
- The remaining data values (\$01 ... \$11) should be spread over the joystick's swing.
- Note: compared with the mouse, the MSbits of the X and Y data bytes are less meaningful, but to stay as compatible as possible with the mouse, the data packet format is kept the same as with the mouse.

3.4.6 Optional byte 3

- This optional byte indicates a Player Control Key (PCK). PCK (6 bit) is applicable (see 3.8 PCK table for values).
- Note: d6 is 0.

3.5 GRAPHIC TABLET/TOUCH SCREEN specification

3.5.1 Data rate

Baudrate: 1200 baud.

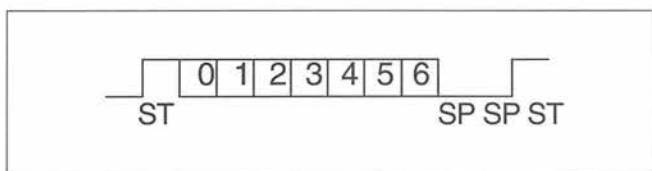
3.5.2 Data flow

During normal operation (RTS asserted) :

- Data is sent to the CD-i player via RXD.
- The 4 byte data packet is sent to the CD-i player continuously when the graphic tablet mouse or pen is on the tablet's active area.
- If the mouse or pen is not on the active area, the graphic tablet should not send any data.
- Every time a button changes state (button1, button2 and/or pendown), a 4 byte data packet is sent to the CD-i player.
- An on going data packet transmission should not be interrupted by a button state change or a mouse/pen position change.

3.5.3 Data format

- 1 startbit (ST).
- 7 databits (LSB first).
- 2 stopbits (SP).



	d6	d5	d4	d3	d2	d1	d0
Byte 0	1	bt1	bt2	X9	X8	X7	X6
Byte 1	0	pd	0	Y9	Y8	Y7	Y6
Byte 2	0	X5	X4	X3	X2	X1	X0
Byte 3	0	Y5	Y4	Y3	Y2	Y1	Y0

- bt1 = button1 : 1 = pressed 0 = not pressed.
- bt2 = button2 : 1 = pressed 0 = not pressed.
- pd = pen down : 1 = mouse/pen on active area 0 = mouse/pen not on active area.

For tablets with switches built into the pen, these switches will be identified as follows:

(FFGB: A VII.2-18) switch1: in tip of pen
switch2: on side of pen.

Switch 1 and switch 2 are equivalent to buttons 1 and 2 respectively.

- X9..X0:
 - 10 bit absolute horizontal position.
 - Outermost left position: \$000.
 - Outermost right position: \$3FF.
- Y9..Y0:
 - 10 bit absolute vertical position.
 - Outermost upper position: \$000.
 - Outermost lower position: \$3FF.

Note: for touch screen bt1 is equal to bt2 and pen down is not relevant.

3.6 CD-i KEYBOARD specification

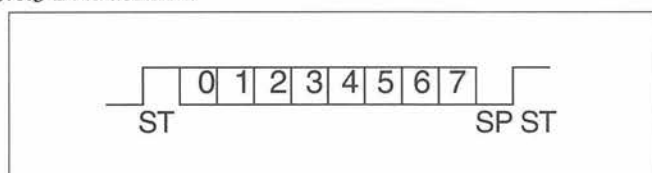
3.6.1 Data rate

Baudrate: 1200 baud.

3.6.2 Data flow

- Data flow during normal operation:
 - Every time there is a keyboard button state change the keyboard sends its current state (last pressed key code or release key code and the special key (*) status) in a 2 byte data packet to the CD-i player via pin 2 of the connector.
 - (*) special keys: Shift, Capslock, Supershift and Control.
 - If there is no button state change, the keyboard should not send any 2 byte keyboard data to the CD-i player.
 - If one or more of the special keys are pressed or released, the keyboard sends the 2 byte data packet with its current special key status, extension bits = 1 and Key-code = \$00.
 - Every time a key is released (no buttons pressed), the keyboard sends the following data:
 - S3-S0: 0000
 - M1-M0: 01
 - K7-K0: 00000000
 This gives the following result: first byte \$82 second \$00.

3.6.3 Data format:



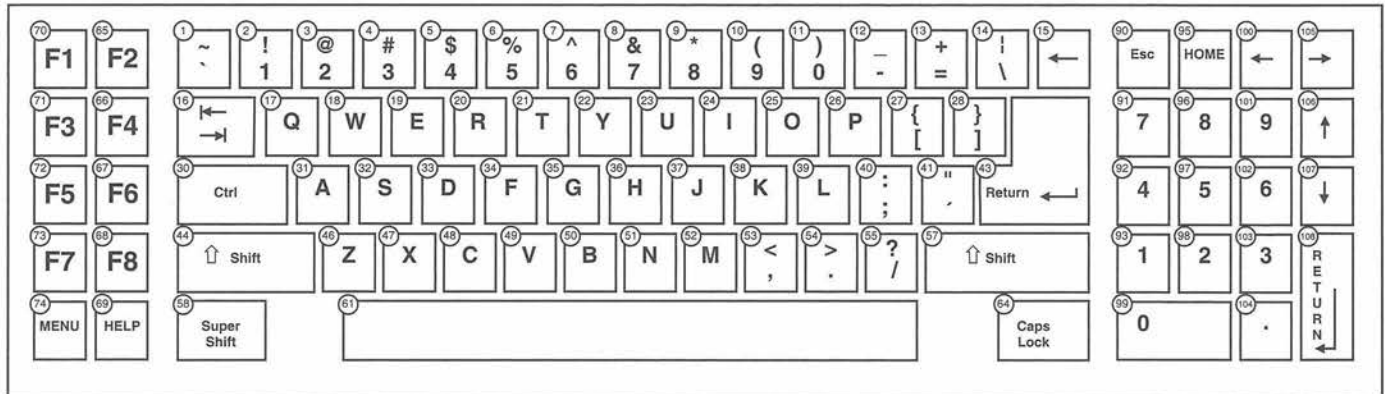
	d7	d6	d5	d4	d3	d2	d1	d0
1st	1	S3	S2	S1	S0	M1	M0	K7
2nd	0	K6	K5	K4	K3	K2	K1	K0

- S0-S3: Status of the special keys :
 - S0: Shift button(s) pressed when set.
 - S1: CapsLock = on when set.
 - S2: Supershift button(s) pressed when set.
 - S3: Control button(s) pressed when set.
- M0-M1: Extension bits :
 - 0 0: Standard Character set (Green Book).
 - 0 1: Character set 1 (if the keycode = \$00 then only special keys are pressed).
 - 1 0: Character set 2 (Future use).
 - 1 1: Character set 3 (Future use).
- K0-K7 Key-code. According to character-set ISO 8859-1 or an extended character-set depending on the M0 and M1 bits.
- Data Range (M&K bits):
 - \$000 - \$0FF: Standard character set (FFGB VI-3).
 - \$100 : Keycode if key released Keycode for special key(s) status.
 - \$101 - \$1FF: Character set 1.
 - \$200 - \$2FF: Character set 2 (Future use).
 - \$300 - \$3FF: Character set 3 (Future use).
- Function CAPSLOCK:
 - Capslock only affects the alphabetical keys.
 - Capslock = off: normal = lower case (\$61-\$7A) shifted= upper case (\$41-\$5A).
 - Capslock = on: normal = upper case (\$41-\$5A) shifted= lower case (\$61-\$7A).

3. INPUT DEVICES

3.6.4 Implementation of a USA English version keyboard

3.6.4.1 Keyboard layout



3.6.4.2 Keyboard character code table USA English version

Keyboard state: CAPSLOCK = off, so Normal = lower case

NR	Normal char	word	char	Shift word	SShift word	Ctrl word	SS+Ctrl word	S+SS word
1	[~]	60	[~]	7E	E0	60	E0	FE
2	[1]	31	[!]	21	B1	31	B1	A1
3	[2]	32	[@]	40	B2	00	80	C0
4	[3]	33	[#]	23	B3	33	B3	A3
5	[4]	34	[\$]	24	B4	34	B4	A4
6	[5]	35	[%]	25	B5	35	B5	A5
7	[6]	36	[^]	5E	B6	1E	9E	DE
8	[7]	37	[&]	26	B7	37	B7	A6
9	[8]	38	[*]	2A	B8	38	B8	AA
10	[9]	39	[()	28	B9	39	B9	A8
11	[0]	30	[)]	29	B0	30	B0	A9
12	[-]	2D	[_]	5F	AD	1F	9F	DF
13	[=]	3D	[+]	2B	BD	3D	BD	AB
14	[N]	5C	[I]	7C	DC	5C	DC	FC
15	delete	7F		0F	0F	1F	0F	0F
16	tab	09		19	19	19	19	19
17	[q]	71	[Q]	51	F1	11	91	D1
18	[w]	77	[W]	57	F7	17	97	D7
19	[e]	65	[E]	45	E5	05	85	C5
20	[r]	72	[R]	52	F2	12	92	D2
21	[t]	74	[T]	54	F4	14	94	D4
22	[y]	79	[Y]	59	F9	19	99	D9
23	[u]	75	[U]	55	F5	15	95	D5
24	[i]	69	[I]	49	E9	09	89	C9
25	[o]	6F	[O]	4F	EF	0F	8F	CF
26	[p]	70	[P]	50	F0	10	90	D0
27	[l]	5B	[L]	7B	DB	5B	DB	FB
28	[l]	5D	[l]	7D	DD	5D	DD	FD
30	ctrl	00		00	00	00	00	00
31	[a]	61	[A]	41	E1	01	81	C1
32	[s]	73	[S]	53	F3	13	93	D3
33	[d]	64	[D]	44	E4	04	84	C4
34	[f]	66	[F]	46	E6	06	86	C6
35	[g]	67	[G]	47	E7	07	87	C7
36	[h]	68	[H]	48	E8	08	88	C8
37	[j]	6A	[J]	4A	EA	0A	8A	CA
38	[k]	6B	[K]	4B	EB	0B	8B	CB
39	[l]	6C	[L]	4C	EC	0C	8C	CC
40	[;]	3B	[;]	3A	BB	3B	BB	BA

41	[']	27	[']	22	A7	27	A7	A2
43	return	0D		0D	0D	0D	0D	0D
44	shift	00		00	00	00	00	00
46	[z]	7A	[Z]	5A	FA	1A	9A	DA
47	[x]	78	[X]	58	F8	18	98	D8
48	[c]	63	[C]	43	E3	03	83	C3
49	[v]	76	[V]	56	F6	16	96	D6
50	[b]	62	[B]	42	E2	02	82	C2
51	[n]	6E	[N]	4E	EE	0E	8E	CE
52	[m]	6D	[M]	4D	ED	0D	8D	CD
53	[.]	2C	[<]	3C	AC	2C	AC	BC
54	[.]	2E	[>]	3E	AE	2E	AE	BE
55	[/]	2F	[?]	3F	AF	2F	AF	BF
57	shift	00		00	00	00	00	00
58	supersh.	00		00	00	00	00	00
61	space	20		20	A0	20	A0	A0
64	capslock	00		00	00	00	00	00
65	F2	81		89	91	99	91	91
66	F4	83		8B	93	9B	93	93
67	F6	85		8D	95	9D	95	95
68	F8	87		8F	97	9F	97	97
69	help	1C		1C	1C	1C	1C	1C
70	F1	80		88	90	98	90	90
71	F3	82		8A	92	9A	92	92
72	F5	84		8C	94	9C	94	94
73	F7	86		8E	96	9E	96	96
74	menu	1D		1D	1D	1D	1D	1D
90	esc	1B		1B	1B	1B	1B	1b
91	[7]	37		37	B7	37	B7	B7
92	[4]	34		34	B4	34	B4	B4
93	[1]	31		31	B1	31	B1	B1
95	home	1E		1E	1E	1E	1E	1E
96	[8]	38		38	B8	38	B8	B8
97	[5]	35		35	B5	35	B5	B5
98	[2]	32		32	B2	32	B2	B2
99	[0]	30		30	B0	30	B0	B0
100	culeft	08		11	15	02	15	15
101	[9]	39		39	B9	39	B9	B9
102	[6]	36		36	B6	36	B6	B6
103	[3]	33		33	B3	33	B3	B3
104	[.]	2E		2E	AE	2E	AE	AE
105	curight	0C		14	18	05	18	18
106	cuup	0B		13	17	04	17	17
107	cudown	0A		12	16	03	16	16
108	return	0D		0D	0D	0D	0D	0D

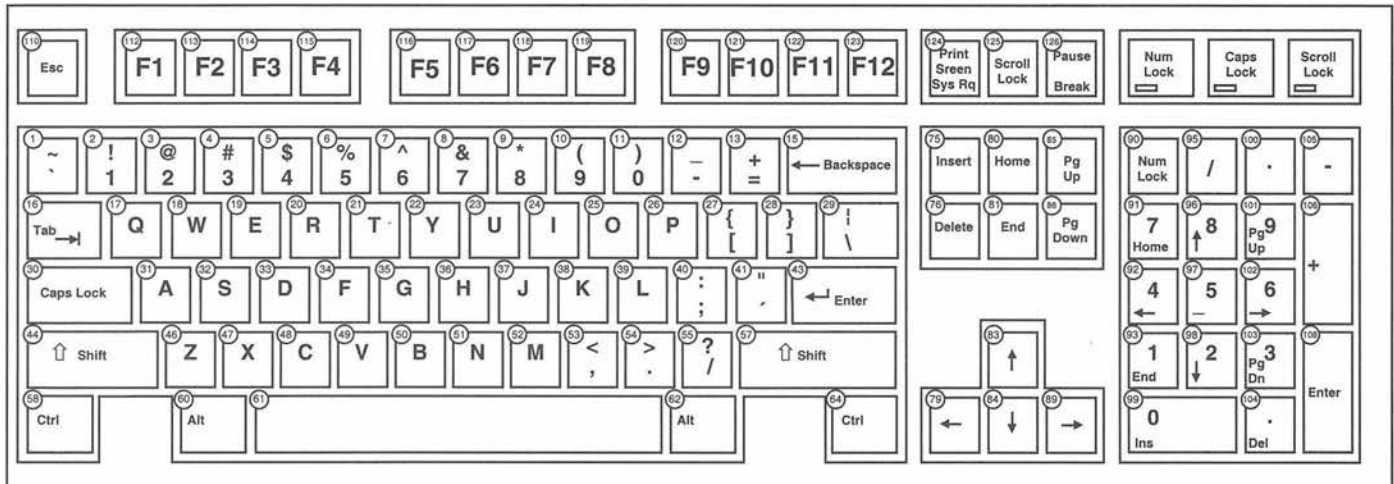
3.6.5 Implementation of an Extended keyboard

The behavior of the extended keyboard is the same as the USA English version keyboard except for following items:

- NUMLOCK function only affects the keys: 91..93, 96..99 and 101..104.
Numlock = off: as described in 3.6.5.2
Numlock = on: as described in 3.6.5.3

- More keys are available:
Print Screen, Sys Req, Scroll Lock, Pause, Break, PC-Delete, Insert, PageUp, PageDown, End, F9, F10, F11 and F12.
- The keyboard can use its own (special) keys to generate some other character codes. These keys are a local matter for the keyboard.
Examples: Numlock, Alt,...

3.6.5.1 Keyboard Layout:



3.6.5.2 Keyboard character code table 1

Keyboard state: CAPSLOCK = off, so Normal = Lower case, NUMLOCK = off

NR	Normal char	word	char	Shift word	SShift word	Ctrl word	SS+Ctrl word	S+SS word
1	[~]	60	[~]	7E	E0	60	E0	FE
2	[1]	31	[!]	21	B1	31	B1	A1
3	[2]	32	[@]	40	B2	00	80	C0
4	[3]	33	[#]	23	B3	33	B3	A3
5	[4]	34	[\$]	24	B4	34	B4	A4
6	[5]	35	[%]	25	B5	35	B5	A5
7	[6]	36	[^]	5E	B6	1E	9E	DE
8	[7]	37	[&]	26	B7	37	B7	A6
9	[8]	38	[*]	2A	B8	38	B8	AA
10	[9]	39	[]	28	B9	39	B9	A8
11	[0]	30	[D]	29	B0	30	B0	A9
12	[-]	2D	[_]	5F	AD	1F	9F	DF
13	[=]	3D	[+]	2B	BD	3D	BD	AB
15	delete	7F		0F	0F	1F	0F	0F
16	tab	09		19	19	19	19	19
17	[q]	71	[Q]	51	F1	11	91	D1
18	[w]	77	[W]	57	F7	17	97	D7
19	[e]	65	[E]	45	E5	05	85	C5
20	[r]	72	[R]	52	F2	12	92	D2
21	[t]	74	[T]	54	F4	14	94	D4
22	[y]	79	[Y]	59	F9	19	99	D9
23	[u]	75	[U]	55	F5	15	95	D5
24	[i]	69	[I]	49	E9	09	89	C9
25	[o]	6F	[O]	4F	EF	0F	8F	CF
26	[p]	70	[P]	50	F0	10	90	D0
27	[l]	5B	[L]	7B	DB	5B	DB	FB
28	[;]	5D	[;]	7D	DD	5D	DD	FD
29	[\]	5C	[]	7C	DC	5C	DC	FC

30	capslock	00		00	00	00	00	00
31	[a]	61	[A]	41	E1	01	81	C1
32	[s]	73	[S]	53	F3	13	93	D3
33	[d]	64	[D]	44	E4	04	84	C4
34	[f]	66	[F]	46	E6	06	86	C6
35	[g]	67	[G]	47	E7	07	87	C7
36	[h]	68	[H]	48	E8	08	88	C8
37	[j]	6A	[J]	4A	EA	0A	8A	CA
38	[k]	6B	[K]	4B	EB	0B	8B	CB
39	[l]	6C	[L]	4C	EC	0C	8C	CC
40	[;]	3B	[;]	3A	BB	3B	BB	BA
41	[']	27	["]	22	A7	27	A7	A2
43	enter	0D		0D	0D	0D	0D	0D
44	shift	00		00	00	00	00	00
46	[z]	7A	[Z]	5A	FA	1A	9A	DA
47	[x]	78	[X]	58	F8	18	98	D8
48	[c]	63	[C]	43	E3	03	83	C3
49	[v]	76	[V]	56	F6	16	96	D6
50	[b]	62	[B]	42	E2	02	82	C2
51	[n]	6E	[N]	5E	EE	1E	9E	DE
52	[m]	6D	[M]	5D	ED	1D	9D	DD
53	[<]	2C	[>]	3C	AC	2C	AC	BC
54	[.]	2E	[>]	3E	AE	2E	AE	BE
55	[/]	2F	[?]	3F	AF	2F	AF	BF
57	shift	00		00	00	00	00	00
58	ctrl	00		00	00	00	00	00
60	supersh.	00		00	00	00	00	00
61	space	20		20	A0	20	A0	A0
62	super sh.	00		00	00	00	00	00
64	ctrl	00		00	00	00	00	00
75	insert	156		156	1D6	156	1D6	1D6
76	pc del.	106		157	1D7	157	1D7	1D7
79	cu left	08		11	15	02	15	15
80	home	1E		1E	1E	1E	1E	1E

3. INPUT DEVICES

81	end	155		155	1D5	155	1D5	1D5
83	cu up	0B		13	17	04	17	17
84	cu down	0A		12	16	03	16	16
85	page up	158		158	1D8	158	1D8	1D8
86	page dwn	159		159	1D9	159	1D9	1D9
89	cu right	0C		14	18	05	18	18
90	numlock			no code send				
91	home	11E	[7]	137	11E	11E	11E	1B7
92	cu left	108	[4]	134	115	102	115	1B4
93	end	165	[1]	131	1E5	1E5	1E5	1B1
95	[/]	12F	[/]	12F	1AF	12F	1AF	1AF
96	cu up	10B	[8]	138	117	104	117	1B8
97	[5]	135	[5]	135	1B5	135	1B5	1B5
98	cu down	10A	[2]	132	116	103	116	1B2
99	insert	168	[0]	130	1E8	130	1E8	1B0
100	[*]	12A	[*]	12A	1AA	12A	1AA	1AA
101	page up	166	[9]	139	1E6	166	1E6	1B9
102	cu right	10C	[6]	136	118	105	118	1B6
103	page dwn	133	[3]	133	1B3	133	1B3	1B3
104	pc del.	106	[.]	12E	106	106	106	1AE
105	[-]	12D	[-]	12D	1AD	12D	1AD	1AD
106	[+]	12B	[+]	12B	1AB	12B	1AB	1AB
108	enter	10D		10D	10D	10D	10D	10D
110	esc	11B	11B	11B	11B	11B		11B
112	F1	80		88	90	98	90	90
113	F2	81		89	91	99	91	91
114	F3	82		8A	92	9A	92	92
115	F4	83		8B	93	9B	93	93
116	F5	84		8C	94	9C	94	94
117	F6	85		8D	95	9D	95	95
118	F7	86		8E	96	9E	96	96
119	F8	87		8F	97	9F	97	97
120	F9	180		188	190	198	190	190
121	F10	181		189	191	199	191	191
122	F11	182		18A	192	19A	192	192
123	F12	183		18B	193	19B	193	193
124	prt sc	140		140	1C0	(*)141	1C1	1C1
125	scrolck	142		142	1C2	142	1C2	1C2
126	pause	143		143	1C3	(**)144	1C4	1C3

(*) Sys Req key.

(**) Break key.

3.6.5.3 Keyboard character code table 2

- Keyboard state: CAPSLOCK = irrelevant
NUMLOCK = on

NR	Normal char	word	char	Shift word	SShift word	Ctrl word	SS+Ctrl word	S+SS word
91	home	137	[7]	11E	1B7	137	1B7	19E
92	cu left	134	[4]	108	1B4	134	1B4	115
93	end	131	[1]	165	1B1	131	1B1	1E5
95	[/]	12F	[/]	12F	1AF	12F	1AF	1AF
96	cu up	138	[8]	10B	1B8	138	1B8	18B
97	[5]	135	[5]	135	1B5	135	1B5	1B5
98	cu down	132	[2]	10A	1B2	132	1B2	18A
99	insert	130	[0]	168	1B0	130	1B0	1E8
100	[*]	12A	[*]	12A	1AA	12A	1AA	1AA
101	page up	139	[9]	166	1B9	139	1B9	1E6
102	cu right	136	[6]	10C	1B6	136	1B6	118
103	pagedwn	133	[3]	187	1B3	133	1B3	1E7
104	pc del.	12E	[.]	169	1AE	12E	1AE	1E9
105	[-]	12D	[-]	12D	1AD	12D	1AD	1AD
106	[+]	12B	[+]	12B	1AB	12B	1AB	1AB
108	enter	10D		10D	10D	10D	10D	10D
110	esc	11B		11B	11B	11B	11B	11B

NOTE

- In the CD-i player the system software will convert the character codes greater than 7F: the 9th bit is cleared.
- Eg.: when pressing on F10 (= code 181) the CD-i application will receive F2 (= code 81) instead.

3.7 AT KEYBOARD SPECIFICATION *

* Not applicable on all models.

3.7.1 Data flow

The AT-keyboard has a synchronous, serial, bidirectional protocol.

The AT-keyboard transmits a scan code to the keyboard controller in the CD-i player whenever a key is pressed. When the key is released, two codes are sent:

- the release code \$F0
- the scan code.

3.7.2 Data format

- 1 startbit (low)
- 8 databits (LSB first)
- 1 paritybit (8 databits + paritybit must be odd)
- stopbit (high).

Clock and data line interpretation:

Clock Data

high	high	= keyboard may send
low	high	= keyboard may not send; CDI 670 busy
high	low	= CDI 670 wants to send a command to the keyboard
low	low	= undefined

For keyboard layout and scan codes, refer to the manual of the AT-keyboard used.

NOTE

- The CD-i system software supports only QUERTY AT-keyboards.
- The CD-i system software passes the same character codes to the CD-i application as defined for the USA English version

3.8 PCK TABLE

3.8.1 PCK

The Player Control Keys (PCK's) which are recognised by the player are listed in the tables below. The PCK (8 bit) is the data which the application will receive on the system calls. The column PCK (6 bit) indicates the corresponding values in the optional byte in the input device data format (see 3.3.4 and 3.4.6).

3.8.2 Definition of 'green' PCK's

PCK (8bit)	PCK (6bit)	Key
\$00	\$00	Key Released
\$80	\$20	Play
\$81	\$21	Stop
\$82	\$22	Pause
\$83	\$23	Next
\$84	\$24	Previous
\$85	\$25	Search Forward
\$86	\$26	Search Reverse
\$87	\$27	Repeat
\$88	\$28	Next Disk
\$89	\$29	Previous Disk
\$8A	\$2A	Still Forward
\$8B	\$2B	Still Reverse
\$8C	\$2C	Slow Forward
\$8D	\$2D	Slow Reverse
\$8E	\$2E	Return
\$90	\$30	0
\$91	\$31	1
\$92	\$32	2
\$93	\$33	3
\$94	\$34	4
\$95	\$35	5
\$96	\$36	6
\$97	\$37	7
\$98	\$38	8
\$99	\$39	9

3.8.3 Definition of 'non green' PCK's

PCK (8bit)	PCK (6bit)	Key
\$00	\$00	Key Released
\$01-0F	\$01-0F	not defined (yet)
\$10	\$10	Menu
\$11	\$11	Standby
\$12	\$12	Mute/Demute
\$1C	\$1C	Help
\$1D	\$1D	CDI/TV
\$1E	\$1E	Volume Up
\$1F	\$1F	Volume Down
\$3B	\$3B	open/close
\$8F	\$2F	not defined (yet)
\$9A-\$9F	\$3A-\$3F	not defined (yet)

NOTE :

The with ** marked remote control functions are functional if the CD-i application is reacting on the PCK keys corresponding with those buttons.

4. ZIP DISK SPECIFICATION

4.1 Introduction

Connecting an external ZIP drive is only available with CD-i players equipped with a parallel port #3.

4.2 Hardware Specifications

INTERNAL DEVICE:

Super I/O FDC37C665 (Registered Name).
Disk controller with digital data separator.

ZIP DRIVE:

IOMEGA Zip 100 parallel port.

4.3 Software Specifications

The firmware drivers support normal access to a zip disk using the '/pcz6_' device name.

If a zip drive is connected and powered during player startup, an entry with device type code 60 for the zip drive is added to the 'csd' file in NVRAM.

The driver supports pc-formatted zip disks. The disks are pre-formatted and may be used without further formatting in the CD-i player.

The access protection features of the zip disks is supported using special GetStat and SetStat calls:

I\$GetStt

ASM CALL: OS9 I\$GetStt
INPUT: d0.w = Path number
d1.w = SS_Protect function code (Hex \$5f)
OUTPUT: d2.b = protection setting of currently mounted zip disk.
value meaning
0 no protection
2 write protected disk
3 password write protected disk
5 password read/write protected disk

ERROR cc = Carry bit set
OUTPUT: d1.w = Appropriate error code

FUNCTION: This call may be used to interrogate the access protection status of a zip disk currently mounted in the zip drive.

POSSIBLE ERRORS: E\$NotRdy, E\$Hardware

I\$SetStt

ASM CALL: OS9 I\$SetStt
INPUT: d0.w = Path number
d1.w = SS_Protect function code (Hex \$5f)
d2.b = required protection setting of currently mounted zip disk.
value meaning
0 no protection
2 write protected disk
3 password write protected disk
5 password read/write protected disk

ERROR cc = Carry bit set
OUTPUT: d1.w = Appropriate error code

FUNCTION: This call may be used to set the access protection parameters of a zip disk currently mounted in the zip drive.

POSSIBLE ERRORS: E\$NotRdy, E\$Hardware, E\$BType, E\$WP

ASM CALL: OS9 I\$SetStt
INPUT: d0.w = Path number
d1.w = SS_Eject function code (Hex \$88)
cc = Carry bit set
OUTPUT: d1.w = Appropriate error code

FUNCTION: This call may be used to eject a disk from the zip drive.

POSSIBLE ERRORS: E\$NotRdy

ASM CALL: OS9 I\$SetStt
INPUT: d0.w = Path number
d1.w = SS_Enable function code (Hex \$98)
cc = Carry bit set
OUTPUT: d1.w = Appropriate error code

FUNCTION: This call may be used to enable the eject command.

POSSIBLE ERRORS: E\$NotRdy

ASM CALL: OS9 I\$SetStt
INPUT: d0.w = Path number
d1.w = SS_Disable function code (Hex \$90)
cc = Carry bit set
OUTPUT: d1.w = Appropriate error code

FUNCTION: This call may be used to disable the eject command (default).

POSSIBLE ERRORS: E\$NotRdy, E\$DIDC

5.1 Introduction

The CSD file is stored in non volatile RAM. This file contains the actual devices. There are two modules that operate in this file during startup: csdinit and config. Csdinit is a so called "kernel customizing module" which is called during the kernel initialization.

5.2 Updating the CSD file depending on devices present.

After startup the config program will first determine which devices are connected to the player. It checks for pointing devices connected to Input 1 and Input 2.

Possible pointing devices are: mouse/trackerball, joystick, graphic tablet, touch screen.

Depending on pointing devices connected to Input 1 and Input 2, the following entries will be put in the CSD :

Connected pointing devices CSD entries for point. dev.

Input 1	Input 2	First entry	Second entry*
Connected	Not conn.	/pt 2	-
Connected	Connected	/pt 2	/pt 1
Not conn.	Not conn.	/pt 2	-
Not conn.	Connected	/pt 1	/pt 2

Most applications react only on one pointing device (also the audio shell of the player) and therefore they will use only the first entry for that device type.

* Only a very limited set of applications request for a second pointing device. They will use also the second entry for that device type. (For example: some of the two player games)

NOTE

- The remote control is always linked to the 'pt2' device. So, the remote control is only active if 'pt2' is an active pointing device.

Config will check for a modem connection on the Serial I/O port and update the CSD entry accordingly.

The DSD for the CD-i keyboard and for the AT keyboard will be added to the CSD file if the corresponding keyboard is found during startup.

The CD-i keyboard can only be connected to Input 2.

For some devices the DSD's are always present in the CSD file: e.g. /t1, /t2, /term. The DSD for the zip 100 drive will be added to the CSD file if the zip drive is connected and powered during player startup.

The following device descriptors are used for the input devices and for the serial devices:

Port	Device Type	Device Descriptor
Input 1	pointing device	/pt2
Input 2	pointing device	/pt1
	CD-i keyboard	/kb1
Port #1	RS232-C serial device	/t1
Port #2	RS232-C serial device	/term
Port #3	parallel device (printer)	/p
Port #4	AT keyboard device	/pckb
Serial I/O port	serial device	/t2

E.g. a pointing device connected to Input 1 and a CD-i keyboard connected to Input 2:

CSD file

```
0:sys:RV="1.1":
1:/cd:
2:/ap:
3:/vid2:LI="525":TV:
4:/nvr:SZ#32:
5:/pt2
6/cdp:
8:mon:AR="4:3":
9:/pipe:
10:/kb1:KG="ALL":LA#0:RP#0:
11:/pck:KG="8":
20:/t2:
20:/term:
20:/t1:
20:/p:
30:/RAM00:SZ#8:AT#150
30:/RAM01:CO#90:AT#700
60:/pcz6_:
90:/mv:RV="CTP3":
90:/ma:RV="CTP3":
```

5. SYSTEM CONFIGURATION

5.3 Load application

When a zip drive is connected, it is possible to load and execute a CD-i application from a zip disk instead of from the CD-i disc mounted in the player. When clicking on the 'CD Play' button or on the 'Play' front panel key, the system will check for a file named '_cdiappl' in NVRAM. If found, it is expected to contain the name of a CD-i application file which must be located on a zip disk mounted in the drive.

The file is opened and the contents loaded into video plane B. The first module in the file is then executed.

NOTE 1:

Creating the file '_cdiappl' is done by:

1. Mount the zip drive on a PC.
2. Use a text editor to create the file.
3. Connect the drive back on the CD-i player.
4. Use the 'Tools' copy option to copy the file from zip disk to player NVRAM.

If the file name starts with the string 'cdi_', the player shell

attempts to set the current directories (command and data) to the CD-i disc root directory before chaining the application. The application will in this case execute as if it was loaded from CD.

If the file name starts with any other string, the player shell attempts to set the current directories to the root directory of the zip disk before chaining the application.

In case of an error while attempting any of the above steps, the player shell will exit and the system will reset.

NOTE 2:

The player shell will only read the first 12 characters from the '_cdiappl' file - this corresponds to the maximum length of a DOS filename.

5.4 Load System Modules

It is possible to provide updates/additions to the OS-9 modules found in the player ROM by using the following method:

Place the updated/new OS-9 modules to load in a file named '_cdimods' in the players NVRAM.

During startup, the player shell will check for the presence of the file in NVRAM and attempt to load all modules found in the file in Video plane A.

NOTE:

The size of the '_cdimods' file is limited to the amount of free space in the NVRAM.

6. PLAYER STATISTICS

A number of player statistics are kept in NV-RAM and are displayed in the 'options' 'copyright' screen of the player shell as a string:

ooooo-ttttt-aaaaa

where:

'ooooo' is the player ON time in hours.
This records the no. of hours the player is powered up.

'ttttt' is the tray open/close counter.
This records the no. of times the tray is opened.

'aaaaa' is the play active time in hours.
This records the no. of hours the player is accessing a disc.
Disc must be loaded and spinning.

7.1 Introduction

This function is only available with CD-i players equipped with serial ports, port #1 and port #2. When the CD-i players are connected to each other by using these serial ports then a network is available. The main feature of this network is to broadcast a message from a "master" CD-i player to all "slave" CD-i player(s). However sending a message to one specific player is not possible.

7.2 Definitions

Master player: A CD-i player in a network which can perform a copy to all the CD-i players in that specific network. There is only one player in a network that can be master. This specific master station can not be changed until all the CD-i players are turned off. The CD-i player that performs the first external copy will be that master player and stays the master player until all CD-i players are turned off.

Slave player: A CD-i player in a network which cannot perform a copy to the network. Only files can be received from a network. All the players in the CD-i network are slave players except the player that performs copy actions to the network (the master player).

CD-i network: A network of CD-i players connected to each other by using a specific connector. There is one master player that performs a copy of files to all the slave players.

OSI-model: OSI-model stands for Open Systems Interconnection reference model. This model is a step into the direction of the international standardisation of different protocols.

7.3 Preconditions

When using the CD-i network facility, the following preconditions must be taken into account:

- When connecting two players to each other, a physical link must be made between port #1 of a player and port #2 of another player.
- The last player of the network may not be connected to the first player of the network. In other words: you cannot create a ring network.
- It is not allowed to disconnect a cable when using the network.
- Different CD-i networks may not be connected. In this situation the identification of the players is not unique and there is a possibility that there is more than one master.
- The player that does the first external copy will be master and stays master until all players are turned off.
- A maximum of 25 CD-i players are recommended.
- Maximum length of the serial connection cable is recommended to be 10m. (33ft.).

When these preconditions are not fulfilled, correct behaviour of the CD-i network cannot be guaranteed.

7.4 Network configuration

For the network, all the CD-i players must be connected through serial ports. This must be done by connecting port #2 of a player with port #1 of another player.

In figure 1 a network configuration of 7 players is shown.

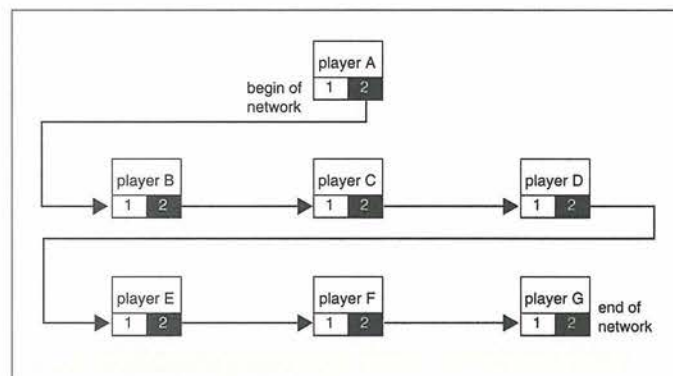


FIGURE 1. CD-i network of 7 players

It is very important to check that these connections are done correctly because the CD-i player does not verify the hardware links.

For network control and also for storage purpose, each player has a unique identification number in a network. This number is not stored in NVRAM. The identification number will be automatically assigned when the network is used for the first time after power on. The assigned number is not visible for the user.

When configuring the network, the preconditions for the network must be fulfilled. A CD-i player within the created network is able to copy a file to an external (remote) player. This player will become master and stays master until the system is turned off.

Configuring the network will pass the following phases:

- Clearing the configurations of all the CD-i stations in the network.
- Assigning identification numbers.
- Broadcast network status to all the CD-i stations.

7.5 CD-i network protocol

The network protocol conforms to the OSI-model standard. Not all the OSI-layers are used within the playershell network program.

7.6 Layer 1: physical link layer

The physical layer concentrates on the communication channel. In our case we need a specific serial connection cable. The connection must have the following layout:

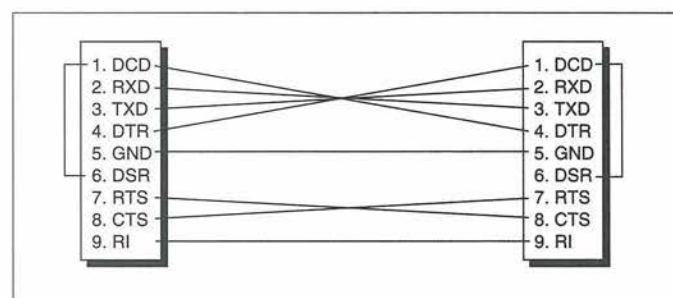


FIGURE 2. CD-i player network connection cable layout

7. GROUPWORKING

7.7 Layer 2: data link layer

The purpose of a data link layer is to provide a transmission-facility which offers a channel to the network layer that is free of transmission errors.

In our case layer 2 has the following functions:

- Error-handling: a simple check-sum is added to the layer 2 datagram. The checksum is established by a 2^c complement of a simple byte arithmetic check modulus 256.
- Message assembling/recognition: create and recognize frame boundaries.

The format for layer 2 consists of:

- Headerbyte: is always FF (hex.)
- Length bytes (3 bytes): size of the layer 3 data.
- Layer 3 data
- Stopbyte: is always 00 (hex.)

7.8 Layer 3: network layer

An important aspect of the network layer is to route messages from source to destination. Within a broadcast environment routing is very simple because unique addressing is not needed. The main goal of the CD-i network is to broadcast data, but for future use unique addressing must be possible. Therefore the network layer for the CD-i network is needed.

The format of a message within the CD-i protocol is always the same. The message always starts with a destination address followed by the address of the source station. When you want to broadcast a message to all stations you must use destination identification 0.

7.9 Layer 4: transport layer

The main function of the transport layer, according to the OSI-model, is to receive data from the session layer and to split this data into smaller parts and offer this to the network layer. In our case there is no need to split the data into smaller parts so we disable the function of the transport layer.

7.10 Layer 5: session layer

The purpose of the session layer is to handle the dialogues between the current station and all the other stations. In our case the session layer has to interpret all received commands and handle these commands correctly.

The format of the session layer consists of four sections: kind of message, command, data and kind of error.

7.11 Summary

The layout of the messages on the different OSI layers is summarized in the figure below.

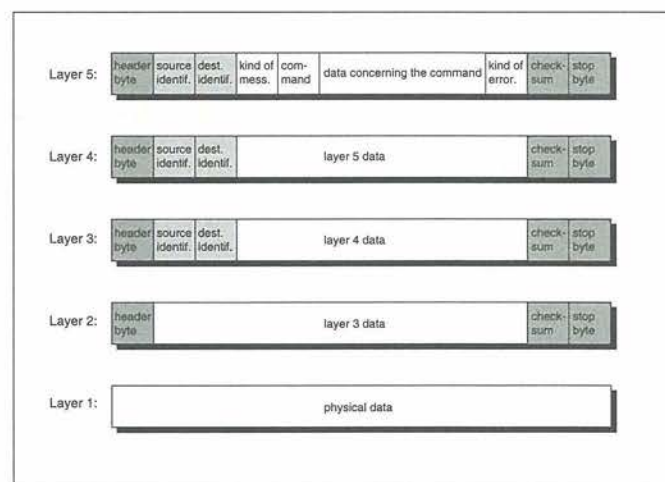


FIGURE 3. Network OSI

8. KEYDISC FUNCTIONS

8.1 Locking features

There are three types of lock functions: disable CD-DA disks, one publisher only (level1) and confidential (level2). The important things here are the "Key Disk". These disks are only supplied by Philips and have a unique code embedded for each customer.

8.2 Lock functions

In normal circumstances -without protection- an application on disk has the same name as its entry in the File Structure Volume Descriptor (FSVD). When starting a CD-i application, the file name and directory path of the application is read from the FSVD, the file loads into memory and the first module in the file starts executing.

The protection mechanism consists in allowing a difference between the FSVD and the applications' file name. A disk especially mastered for level 2 protection contains two applications: one application with a standard name and one application with a postfix added to the same standard name. A normal player will read the standard filename in the FSVD and play the 'standard' application. Only a player able to filter out the difference between both names will play the application with the adapted application. The standard application on the disk will thus play on all players, except on the player the disk is locked to (when locking is enabled). This application could be used to inform the user that the disk he just inserted is intended for someone else's eyes. The difference between both names is created by adding a postfix to the normal name:

Adapted name=
"FSVD name"
& "company dependent part"
& "variable code"

A protected file name on disk with 'cdi_show' as FSVD-entry would become:

Adapted name=
"cdi_show"
& "_LCK"
& "0001",

resulting in 'cdi_show_LCK0001' as filename on disk.

The variable code in the adapted file name can be used in many different ways: as identification number of the application, as chapter number when the total package has one application per chapter, as security level,... The player holds the 'company dependent part' together with a list of four digit numbers (equivalent to the adapted application names on disk) and only plays the disk that have a number from this list in their adapted application name. This content of the list depends on the applications the player may or may not run. The confidential application will not play when its name does not match one of the entries in the list.

An example:

The company Bigdeal distributed the product information to their dealers on CD-i disks (mastered with TCM in mind). The complete set, divided over several disks, consists of a documentation part, a sales training, technical instruction, etcetera. Bigdeal also ordered a keydisk with PRO1 as company dependent part. The player of the sales people was locked to the sales training and wouldn't play any of the disks they received except the sales disk. Because the sales people are humans too, they inserted the technical disk in their player. The player responded this action with a message on the screen: "The technical application will not run on this player". Suppose the adapted name of the sales application was 'salesPRO10004',

then the entry in the postfix file would be '0004'. The player of the technical staff may access the documentation (with docPRO10002 as name) and the technical disk (with techPRO10007 as name). The contents of the postfix file in their player would be '0002' and '0007'.

8.3 Secondary locking function

Unfortunately, the locking features just described are only applicable to CD-i disks. In some cases the locking features can't be implemented on a CD-i title due to limitations in the pre-mastering step. The locking method can not be implemented on Video CD's or Photo-CD's. To overcome this drawback, a secondary protection mechanism (level 1) is implemented in the player.

The mechanism is similar to the previous described system but uses other data to check permissions. When locking is enabled and there is a postfix file in memory -with at least one entry- the inserted disk is searched for adapted filenames. In case no names are found, the player checks the four first characters of the 'Publishers string' in the FSVD on disk. When they match with the code in the postfix file, the disk will play. Else, the disk is ejected and the customer specific message appears on the screen.

The major difference between both locking systems is that the level 1 protection connects the disk(s) to a particular player while the level 2 protection locks the player to a particular disk. In the first case the disk(s) will not play on other players and the player won't play other disks. In the second case, the locked player will only play the protected disks, with the major difference that these disks can play on any other player.

8.4 CD-DA lockout

The CD-DA disable function is rather straight forward: when keydisk locking is enabled, the player refuses to play audio disks.

The following table gives an overview of the different disk-player combinations possible:

	Normal player	CD-i in locked mode
Normal disk	Will play	Will not play
CD-DA disk	Will play	Will not play
One publisher only (level 1)	Will play	Will play (*)
Confidential disk (level 2)	Will not play / play different	Will play (*)

(*) if lock strings are matching

8.5 Other key disks functions

Depending on the customers needs:

- Option access protection.
- Start-up messages.
- Pointing device start-up (see also 3.2.4).

NOTE

- The Load System Modules (see also 5.4) feature enables a great variety of custom developed features/drivers.

